

Geophysical Research Letters[®]



RESEARCH LETTER

10.1029/2026GL122098

Key Points:

- PACE hyperspectral reflectance shows consistent differences between low and high microcystin concentrations in Western Lake Erie
- Pigment-band spectral shape metrics enhance screening for microcystin exceedance beyond just reflectance data alone
- A random forest classifier detects microcystin exceedance with high accuracy on an independent test data set

Supporting Information:

Supporting Information may be found in the online version of this article.

Correspondence to:

A. Ebtehaj,
ebtehaj@umn.edu

Citation:

Karunakaran, D., Ebtehaj, A., Clark Burnette, L., Hondzo, M., & Olmanson, L. (2026). Initial assessment of the detectability of cyanotoxin occurrence using PACE satellite observations. *Geophysical Research Letters*, 53, e2026GL122098. <https://doi.org/10.1029/2026GL122098>

Received 31 JAN 2026

Accepted 17 MAY 2026

Author Contributions:

Conceptualization: Denesh Karunakaran, Ardeshtir Ebtehaj, Lucas Clark Burnette, Miki Hondzo, Leif Olmanson

Data curation: Denesh Karunakaran, Ardeshtir Ebtehaj, Lucas Clark Burnette

Formal analysis: Denesh Karunakaran, Ardeshtir Ebtehaj, Lucas Clark Burnette, Miki Hondzo, Leif Olmanson

Funding acquisition: Ardeshtir Ebtehaj, Miki Hondzo, Leif Olmanson

Investigation: Denesh Karunakaran, Ardeshtir Ebtehaj, Lucas Clark Burnette, Miki Hondzo, Leif Olmanson

© 2026. The Author(s).

This is an open access article under the terms of the [Creative Commons Attribution-NonCommercial-NoDerivs License](#), which permits use and distribution in any medium, provided the original work is properly cited, the use is non-commercial and no modifications or adaptations are made.

Initial Assessment of the Detectability of Cyanotoxin Occurrence Using PACE Satellite Observations

Denesh Karunakaran¹ , Ardeshtir Ebtehaj¹ , Lucas Clark Burnette¹, Miki Hondzo¹ , and Leif Olmanson²

¹Saint Anthony Falls Laboratory, Department of Civil, Environmental, and Geo-Engineering, University of Minnesota Twin Cities, Minneapolis, MN, USA, ²Department of Forest Resources, University of Minnesota, Saint Paul, MN, USA

Abstract Microcystin is a major public health hazard during freshwater cyanobacterial blooms, yet its remote detection remains challenging because it lacks any direct optical absorption features. We demonstrate that hyperspectral observations from NASA's PACE Ocean Color Instrument (OCI) can be used to detect elevated microcystin concentrations by exploiting bloom-related optical signatures. Two years of NOAA in situ microcystin measurements from the Western Lake Erie Basin are analyzed using bootstrapped Welch's *t*-tests to identify OCI wavelengths sensitive to high-microcystin conditions. These bands and derived spectral-shape metrics are then used to train a nested cross-validated random forest classifier, achieving 89.5% accuracy despite limited training data. The results reproduce expected spatial and seasonal patterns of toxic blooms and highlight the dominant predictive role of spectral-shape metrics in the pigment absorption and red-edge regions.

Plain Language Summary Toxic cyanobacterial blooms can produce microcystin, a harmful and potent toxin that poses a threat to drinking water and public health. Detecting this toxin from satellite data is difficult because it lacks a clear optical signal. In this study, we combine field measurements from the western Lake Erie basin with new hyperspectral satellite imagery to identify indirect optical features and patterns associated with high toxin levels. We find that specific wavelengths, especially in the red and red-edge regions, are strongly associated with elevated microcystin concentrations. These results show that a new generation of hyperspectral spaceborne sensing systems can help identify areas with elevated risks of toxic algal blooms.

1. Introduction

Harmful algal blooms (HABs) are dense accumulations of algal biomass, often dominated by toxin-producing cyanobacteria (Huisman et al., 2018). These blooms degrade water quality (Paerl & Otten, 2013; Sukenik & Kaplan, 2021) and threaten human and ecosystem health through cyanotoxins, particularly microcystin (MC) (Carmichael, 2001). MC is a hepatotoxin (Bouaïcha et al., 2019) whose production is regulated by both bloom genetics (Nishizawa et al., 1999) and environmental conditions, including temperature, nutrients, and light (Harke et al., 2016; Kaplan et al., 2012; Wiedner et al., 2003). Although bloom intensification is widely linked to nutrient enrichment and warm surface waters (Feng et al., 2024; Ho et al., 2019), increased biomass is not consistently correlated with MC occurrence (Harke et al., 2016), which limits inference from biomass alone and motivates scalable synoptic screening approaches.

Remote sensing enables large-scale HAB monitoring; however, MC lacks direct optical absorption features (Stumpf et al., 2016). Toxic cyanobacterial blooms modify water-leaving reflectance via pigment absorption, such as phycocyanin (PC) and chlorophyll *a* (Chl *a*), enhanced particulate backscattering, and changes in dissolved organic matter (Hunter et al., 2016; Matthews, 2011; Moore et al., 2017; Simis et al., 2005). These bio-optical responses provide a basis for proxy-based screening of elevated MC risk (Stumpf et al., 2016; Zhang et al., 2017).

While recent efforts have successfully evaluated a multi-spectral satellite-based algorithm against coincident MC measurements (Mishra et al., 2021), the coarse spectral resolution of most operational multi-spectral sensors limits capturing diagnostic information of spectral shapes in optically complex inland waters (Dierssen et al., 2021; Wang & Qin, 2025; Zolfaghari et al., 2021). NASA's Plankton, Aerosol, Cloud, ocean Ecosystem (PACE) mission provides routine, swath-based hyperspectral ocean-color observations from the Ocean Color Instrument (OCI), spanning 315–895 nm with a spectral bandwidth of ~5 nm, sampling intervals of 1.25 and 2.5 nm, and near-daily global coverage at ~1.2 km grids at nadir (Dierssen et al., 2023; Werdell et al., 2019).

Methodology: Denesh Karunakaran, Ardeshir Ebtehaj, Lucas Clark Burnette, Miki Hondzo, Leif Olmanson
Project administration: Denesh Karunakaran, Ardeshir Ebtehaj, Miki Hondzo, Leif Olmanson
Resources: Denesh Karunakaran, Ardeshir Ebtehaj, Lucas Clark Burnette, Miki Hondzo, Leif Olmanson
Supervision: Denesh Karunakaran, Ardeshir Ebtehaj, Miki Hondzo, Leif Olmanson
Validation: Denesh Karunakaran, Ardeshir Ebtehaj
Visualization: Denesh Karunakaran, Ardeshir Ebtehaj, Lucas Clark Burnette
Writing – original draft: Denesh Karunakaran, Ardeshir Ebtehaj, Lucas Clark Burnette, Miki Hondzo
Writing – review & editing: Denesh Karunakaran, Ardeshir Ebtehaj, Lucas Clark Burnette, Miki Hondzo, Leif Olmanson

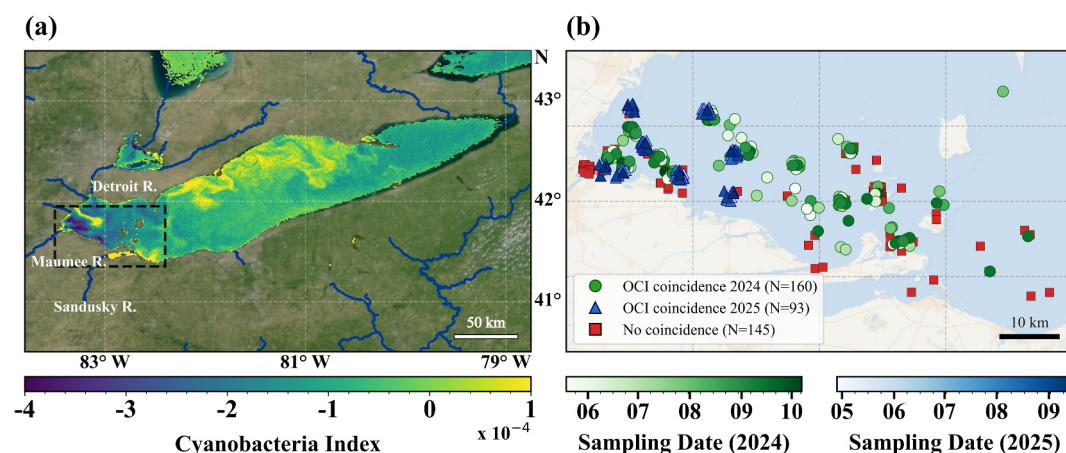


Figure 1. (a) Cyanobacteria Index derived from PACE Ocean Color Instrument (OCI) reflectance over Lake Erie on 14 September 2025 showing low (CI < 0) and high algal biomass (CI > 0), and (b) locations of in situ sampling sites in 2024 (circles) and 2025 (triangles). Sampling sites without valid OCI matchups are shown as squares, and marker colors show sampling month.

This study evaluates OCI's capability for proxy-based screening of elevated MC conditions in large inland and coastal waters. We combine 2 years of in situ MC measurements from the Western Lake Erie Basin (WLEB) with near-coincident OCI hyperspectral reflectance to quantify spectral differences between high- and low-MC conditions. Spectral separability is assessed using a bootstrapped, band-wise Welch's *t*-test with FDR control (Benjamini & Hochberg, 1995; Efron & Tibshirani, 1994). Significant wavelengths and continuum-removed pigment band-shape metrics are then used to train a random forest classifier within a nested cross-validation framework to mitigate overfitting under limited sample sizes. Together, these results provide proof of concept that PACE OCI can enable scalable spaceborne monitoring of elevated MC in optically complex inland waters with high accuracy.

2. Data and Methods

2.1. Study Area and In Situ Measurements

Lake Erie is the shallowest of the Laurentian Great Lakes, with an average depth of 7.4 m in its western basin (Millie et al., 2009). The WLEB receives nutrient-rich inflows from the Detroit and Maumee Rivers (International Joint Commission (IJC), 2014). Its short residence time of ~51 days and frequent sediment resuspension enhance nutrient recycling, and increase susceptibility to HABs (Binding et al., 2010; Millie et al., 2009). Since the 1970s, the WLEB has experienced more frequent and intense HABs (Makarewicz, 1993) and serves as a key testbed for cyanobacterial spectral studies, particularly *Microcystis* (Stumpf et al., 2016; Vincent et al., 2004; Wynne et al., 2010).

In situ water-quality data are obtained from NOAA's weekly monitoring program archived at the National Centers for Environmental Information (NCEI) (Chaffin, 2023; Cooperative Institute for Great Lakes Research et al., 2019). The data set includes 306 samples (June–October 2024) and 150 samples (May–September 2025) collected across the WLEB. Measurements comprise total MC, pigments, nutrients, and ancillary variables (see in Supporting Information S1). Figure 1a shows the study area with the OCI-derived Cyanobacteria Index (CI) (Wynne et al., 2008), while Figure 1b depicts the spatiotemporal distribution of in situ sampling locations.

2.2. PACE OCI Observations and Near-Coincident Matchups

The OCI hyperspectral surface reflectance ρ_λ is obtained from NASA's Ocean Biology Distributed Active Archive Center (Caplan et al., 2025). We use the level-2 regional surface reflectance product (V.3.1) at ~1.2 km spatial resolution at nadir across 122 spectral bands spanning 346–895 nm (NASA Ocean Biology Processing Group, 2025). Because cloud cover, adjacency effects, and elevated aerosol loading often limit same-day OCI coverage during the bloom season (Sayers et al., 2019), we form the matchups using a 3-day temporal window, defined as ± 1 day from each in situ measurement (Bailey & Werdell, 2006; Olmanson et al., 2008, 2011).

For each in situ sample, we select the nearest OCI pixel that passes the standard high-quality screening flags (Bailey & Werdell, 2006) within 3 km. If the nearest pixel fails quality control or contains missing values, the next closest valid pixel within 3 km is used. When multiple valid OCI observations occur within the window, the spectra are averaged to form a single matchup, without any outlier removal.

2.3. Methods

2.3.1. Band Metrics

Band-metric analyses are well established in terrestrial spectroscopy but are less explored for inland HAB monitoring. Here, we compute continuum-removed band center (BC), band depth (BD), and band area (BA). For each wavelength window $[\lambda_L, \lambda_R]$, the continuum $C(\lambda)$ is defined as the upper convex hull between the shoulder wavelengths, and the normalized spectrum is obtained as $R_n(\lambda) = R(\lambda)/C(\lambda)$ (Clark & Roush, 1984; Kokaly & Clark, 1999). We apply this to two pigment-informed windows: the PC window (550–650 nm) and the Chl *a* red-edge (CRE) window (650–700 nm). BC is the wavelength that minimizes $R_n(\lambda_{\min})$, $BD = 1 - R_n(\lambda_{\min})$, and $BA = \int_{\lambda_L}^{\lambda_R} [1 - R_n(\lambda)] d\lambda$. An example is provided in Figure S1 of the Supporting Information S1 Document.

2.3.2. Bootstrapped Hypothesis Testing

We consider a null hypothesis H_0 that OCI spectral reflectance does not differ between low- and high-MC conditions. Spectra are divided into two groups: (a) $MC < 1 \mu\text{g L}^{-1}$ (low) and (b) $MC \geq 1 \mu\text{g L}^{-1}$ (high), using the World Health Organization (WHO) guideline threshold (WHO, 2003). We frame H_0 at each wavelength as equality of group-mean reflectance, $\mu_l(\lambda) = \mu_h(\lambda)$, and test it using a bootstrap implementation of Welch's two-sample *t*-test (Efron & Tibshirani, 1994). Please see the detailed derivation in Text S2 of Supporting Information S1. Because we test H_0 across wavelengths independently, there is a risk of inflated type-I error. Therefore, we adjust *p*-values with the Benjamini–Hochberg FDR method (Benjamini & Hochberg, 1995) using $\alpha = 0.01$. Only bands that remain statistically significant after FDR correction are included in subsequent predictive analysis.

2.3.3. Random Forest Classification

A random forest (RF) classifier (Breiman, 2001) is trained with statistically significant spectral bands and computed band metrics. The response variable is a binary label, with zero for $MC < 1 \mu\text{g L}^{-1}$ and one otherwise. The data set is partitioned into 70 (30)% for training (testing), with class imbalance preserved. Given the limited sample size, a nested 5-fold cross-validation (CV) framework (Cawley & Talbot, 2010; Kim, 2009) is adopted to reduce overfitting, avoid optimistic bias, and ensure robust generalization (see Figure S2 of the Supporting Information S1). Nested CV uses outer and inner loops to separate hyperparameter tuning from model evaluation.

Inner-loop hyperparameter tuning is performed via 200 uniform random draws from the hyperparameter search space, including the number of trees (900–1300), maximum depth (9–11), feature subsampling fractions (0.3–0.5), minimum samples for splitting at internal (8–12) and leaf (6–10) nodes, as well as split criteria (Gini, cross-entropy). To mitigate the effects of class imbalance on RF performance (Davis & Goadrich, 2006; Saito & Rehmsmeier, 2015), optimal hyperparameters are selected by maximizing the area under the precision–recall curve (AUPRC). Moreover, outer-loop class probabilities are used to determine an optimal decision threshold that maximizes the F1-score, rather than relying on the default majority-vote rule. A feature-importance analysis is conducted to identify spectral bands most strongly associated with elevated MC. Final hyperparameters are summarized in Table S1 of Supporting Information S1, and the full RF classification workflow is described in Text S3 of the Supporting Information S1.

3. Results and Discussion

3.1. Bloom Toxicity Dynamics and Reflectance

Prior to predictive analysis, we summarize the matchup data set and its bio-optical context. Figure 2a presents the in situ MC time series for the 2024 and 2025 bloom seasons for four range groups. MC concentrations remain near background levels ($\leq 0.15 \mu\text{g L}^{-1}$) through June to mid-July, rise rapidly in late July, and reach a mid-August maximum, exceeding $6 \mu\text{g L}^{-1}$. In 2024, a secondary pulse occurs in early September, whereas 2025 is dominated by a single late-August peak. The MC distribution across all matchups (Figure 2b) is imbalanced, ~185 samples

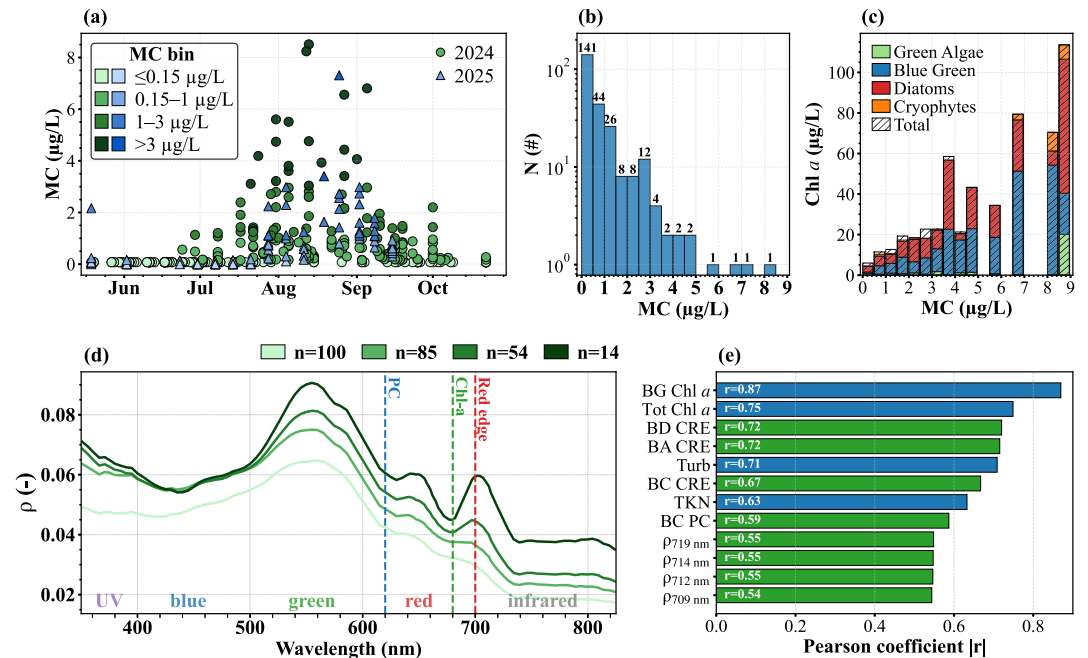


Figure 2. (a) In situ MC time series for the bloom seasons of 2024 (circles) and 2025 (triangles) for four concentration groups, (b) frequency distribution of MC for all valid in situ-OCI matchups, (c) median Chl *a* concentration for different MC values and group-specific algal community, (d) mean surface reflectance spectra for the defined MC groups in (a) with *n* sample size in each group, and (e) Pearson correlation coefficients between MC and the top-correlated variables (all *p* < 0.05) with blue (green) bars denote in situ (Ocean Color Instrument) variables.

fall below $1 \mu\text{g L}^{-1}$ and 68 exceed this threshold, with four samples $>5 \mu\text{g L}^{-1}$. Pigment composition varies across MC levels (Figure 2c), with diatoms and blue-green algae dominating the Chl *a* pool and minor contributions from green algae and cryptophytes.

The mean reflectance spectra per MC range groups are shown in Figure 2d. The results show that the reflectance increases systematically across the visible to near-infrared (VNIR) range with MC, consistent with enhanced particulate backscattering under intensified bloom conditions. The dominant enhancements occur near the green peak at $\sim 550 \text{ nm}$ and in the red-edge/NIR shoulder at $\sim 709 \text{ nm}$, consistent with established spectral signatures of dense surface blooms (Dekker et al., 1993; Stumpf et al., 2012). The red-region troughs around 620 and 680 nm deepen with MC, reflecting the strengthening of Chl *a* and PC absorption (Gitelson et al., 1993; Yacobi et al., 2015).

Figure 2e demonstrates that the blue-green Chl *a* shows the highest positive correlation with MC at 0.87, followed by total Chl *a* at 0.75, indicating that toxin levels scale most directly with cyanobacterial biomass rather than that of total phytoplankton biomass. The OCI-derived BD and BA in the CRE window exhibit correlations of ~ 0.72 , highlighting the predictive value of red and red-edge features, which are enhanced by increasing pigment absorption and particle scattering (Binding et al., 2013; Gitelson et al., 1993; Gower et al., 2008).

In situ turbidity (Turb) and total Kjeldahl nitrogen also show correlation of about 0.71 and 0.63 with MC, reflecting the particle and nutrient-rich environmental characteristic of mature cyanobacterial blooms (Francy et al., 2016; Paerl & Otten, 2013). The next-strongest correlations are the BC metrics from both the CRE and PC windows, with coefficients of 0.67 and 0.59, respectively. These relatively high correlations partly reflect redward shifts of intracellular pigment absorption features (Gons, 1999; Janatian et al., 2025), enhanced PC absorption near 620 nm (Simis et al., 2005), and particulate scattering that modifies the red-edge slope (Gower et al., 2008).

At the band level, wavelengths 709–719 nm near the red-edge/NIR shoulder show the strongest correlations with MC, supporting consistent bloom-related spectral sensitivity (Gitelson et al., 1993; Gower et al., 2008). These

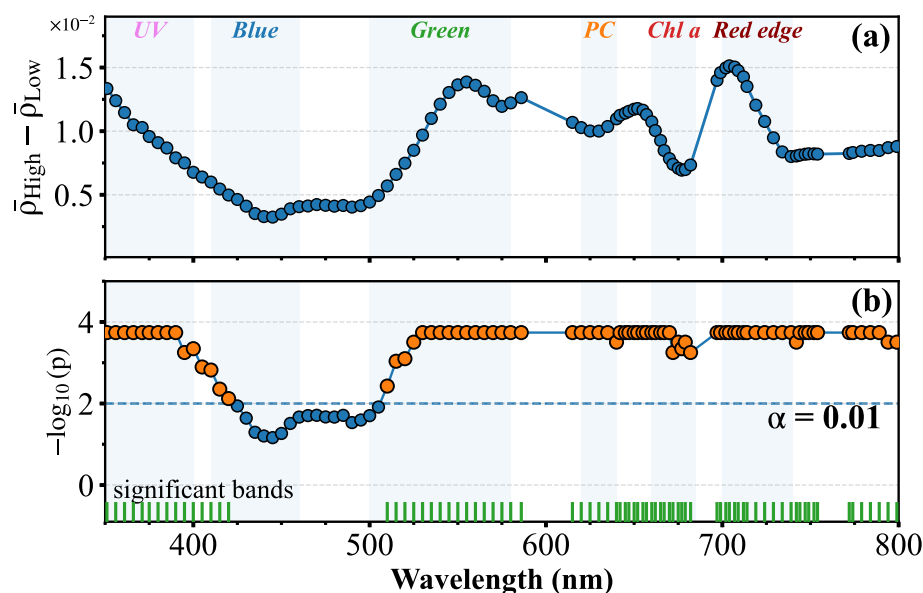


Figure 3. (a) Differences of between-group mean reflectance and (b) FDR-adjusted p -values from a bootstrapped Welch's t -test, where the dashed line delineates the significance level $\alpha = 0.01$. Orange markers denote significant wavelengths, and green ticks denote bands retained for subsequent analysis.

correlations are not statistically independent but provide insight into the relative importance of different predictors, with OCI-derived band metrics exhibiting correlations comparable to key in situ variables.

3.2. Significant Spectral Signatures

Figure 3a shows the difference between the mean reflectance spectra of high- and low-toxin groups: $\bar{\rho}_{\text{High}} - \bar{\rho}_{\text{Low}}$. The largest group-mean differences occur in the ultraviolet (UV), green, and red-edge regions, while differences in the blue region are comparatively small. Hypothesis-testing results (Figure 3b) identify wavelengths at which these differences are statistically significant, indicating that reflectance differs between low- and high-toxin waters across nearly the entire VNIR spectrum. The only exception is a narrow blue interval ≈ 425 –505 nm, where the null hypothesis cannot be rejected.

The lack of statistical significance in the 425–505 nm range is consistent with prior studies showing that, in optically complex waters, blue to blue-green reflectance cannot be uniquely diagnostic due to overlapping absorption by phytoplankton pigments, detritus, and CDOM (Sauer et al., 2012). Discriminatory information, therefore, emerges more clearly at longer wavelengths, where red, red-edge, and near-infrared reflectance respond more distinctly to bloom dynamics. In this data set, significant mean differences across the green–red and red-edge regions are consistent with absorption by Chl a and PC (Gitelson et al., 1993; Kutser, 2004; Simis et al., 2005), along with enhanced particulate scattering that increases red-edge reflectance under high-biomass concentration (Gower et al., 2008; Matthews, 2015). Accordingly, wavelengths that are not statistically significant at $\alpha = 0.01$ are excluded from subsequent predictive analyses.

3.3. Detection Accuracy and Feature Importance

Using the statistically significant bands and spectral metrics, the trained RF successfully distinguishes between low- and high-MC waters. On the independent 30% test set, the model achieves an AUPRC of 0.841 and an area under the receiver operating characteristic curve (AUROC) of 0.945 (Figure 4a). Performance remains consistently high across outer cross-validation folds (i.e., AUPRC = 0.86 ± 0.08 , AUROC = 0.93 ± 0.04), with modest variability, indicating stable performance across independent partitions. The close agreement between cross-validation and test performance also suggests minimal overfitting and robust generalization. As a further sanity check, a label-permutation test yields near-chance AUPRC and AUROC values (Figure 4a), confirming that the observed performance arises from inherent predictive structure rather than data leakage or artifacts.

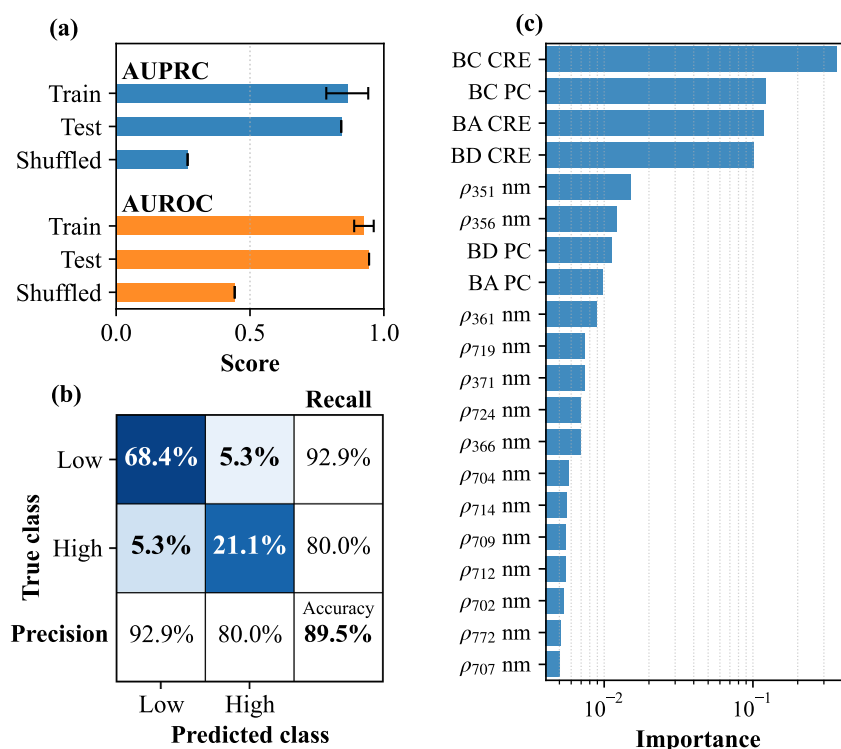


Figure 4. (a) AUPRC and AUROC across outer cross-validation folds for the training set (Train), compared with the independent test set (Test) and a label-shuffled baseline (Shuffled). (b) Confusion matrix for the independent test set at the optimal probability threshold ($p = 0.51$). (c) Bootstrap feature importance for the top 20 predictors, including Ocean Color Instrument reflectance bands and band-metric features. PC and CRE denote the phycocyanin and chlorophyll-*a* red-edge windows, where BC, BD, and BA denote band center, depth, and area.

The confusion matrix (Figure 4b) shows an accuracy of 89.5% and an F1 score of 0.80 at an optimized probability threshold of 0.51. Errors are distributed comparably across classes, suggesting a well-calibrated threshold with balanced false positives (false alarms) and false negatives (misses). Most low-MC samples are correctly identified, with a recall and precision of 92.9%, indicating high reliability under low-toxicity conditions. High-MC waters are also detected with substantial skill, with recall and precision of 80.0%, indicating some missed high-MC events.

Feature-importance results (Figure 4c) show that derived band-metric predictors contribute substantially beyond individual wavebands. Metrics capturing spectral-shape variations in the CRE and PC regions rank highest, particularly band-center metrics that reflect redward shifts in CRE and PC absorption features with increasing toxin concentration, consistent with the strong positive correlations in Figure 2e. Band area and depth likely capture spectral broadening and deepening as cyanobacterial biomass and toxin levels increase. Several red-edge bands, especially near 720 nm, also rank highly, consistent with enhanced particulate backscattering in cyanobacteria-dominated, high-biomass waters (Gower et al., 2008; Matthews, 2011).

Across the UV range 350–375 nm, the reflectance features are also ranked relatively important. The group-mean spectra (Figure 2d) show lower reflectance in this region for the lower-toxin classes, indicating stronger short-wavelength absorption. Reflectance variability in this range is strongly influenced by CDOM and non-algal particles (NAP), including fine clays and inorganic minerals, which increase absorption and suppress near-UV reflectance (Mannino et al., 2019; Mobley, 2022). Elevated CDOM and NAP are consistent with fresher inflow conditions and increased allochthonous inputs, which are more commonly associated with lower-bloom states. Because most satellite sensors provide limited coverage in the 350–375 nm range, future work should further evaluate the influence of this proxy.

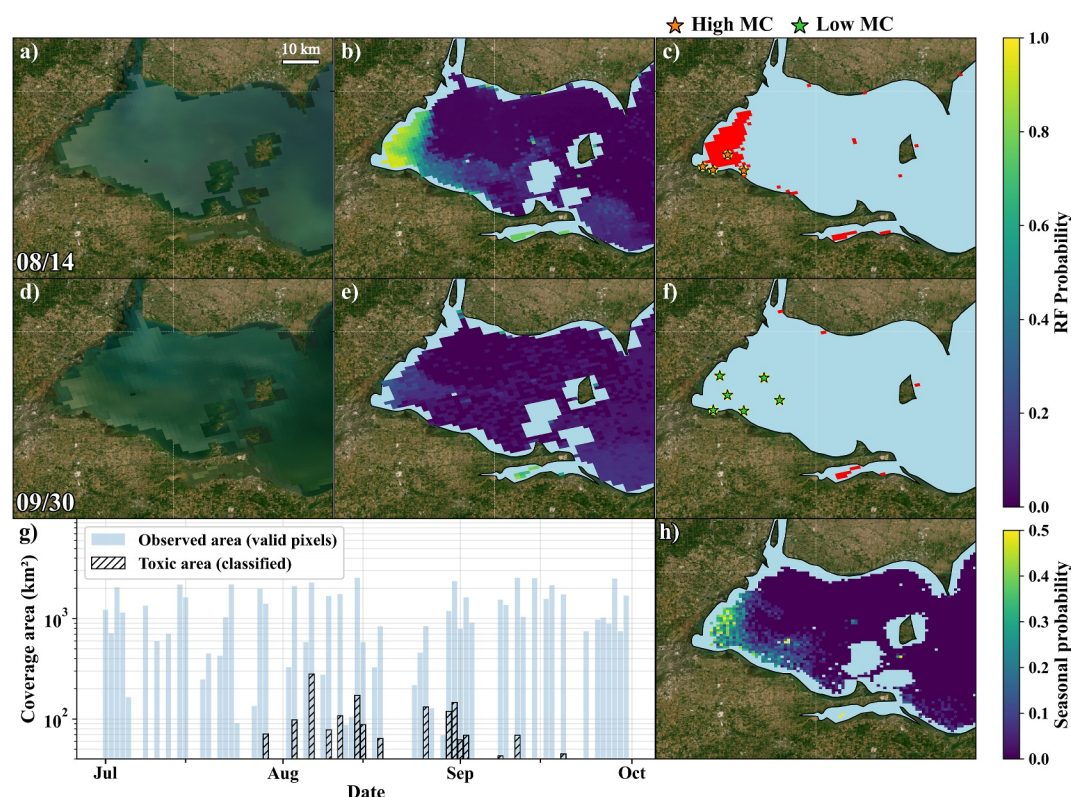


Figure 5. The performance of the predictive model applied to two Ocean Color Instrument (OCI) overpasses. (a, d) True-color imagery on 14 August 2025 (peak bloom) and 30 September 2025 (late season). (b, e) Predicted probability of high-MC values and (c, f) predicted class labels with the withheld in situ MC class are shown with color-coded star markers. (g) Seasonal time series of the fraction of valid (clear-sky) OCI pixels and those classified as high MC from 1 July to 30 September 2025. (h) Seasonal frequency of high-MC classifications over the same period, based on all clear-sky overpasses.

3.4. MC Detection Using PACE Observations

The RF model is applied to two OCI overpasses representing peak and late-season bloom conditions to evaluate performance against held-out in situ measurements (Figure 5). During the peak-bloom overpass on 14 August 2025 (Figures 5a–5c), true-color imagery shows turbid blooms across the Maumee and Sandusky Bays. Predicted MC occurrence probabilities are elevated in these nearshore waters, increasing sharply from approximately 0.4 offshore to 0.6 near the shoreline. The resulting high-MC class map (Figure 5c), validated by independent in situ observations (orange stars), confirms the detection skill of the trained model. All four in situ samples indicating high-MC conditions with concentrations ranging from 1.12 to 6.27 $\mu\text{g L}^{-1}$ are correctly classified. During the late-season overpass on 30 September 2025 (Figures 5d–5f), the model assigns uniformly low-MC occurrence probabilities across most of the basin, including nearshore regions which are typically prone to bloom development. The probability fields and low-MC class map are consistent with the five available in situ observations, not used in training, which report MC concentrations ranging from 0.10 to 0.81 $\mu\text{g L}^{-1}$. Overall, despite its pixel-scale training, the model produces spatially coherent patterns consistent with expected toxic bloom occurrence.

To further assess model validity given the small training sample size, we evaluate seasonal performance against known space-time climatology. The trained RF is applied to all OCI overpasses between 1 July and 30 September 2025. For each overpass, we first compute the area of available clear-sky pixels over the western basin and then report the fraction of high-MC areas (Figure 5g). The seasonal spatial probability of detected high-MC pixels is also shown in Figure 5h. Despite the limited training data, the results yield two key observations. First, the time series shows that high-MC areas emerge in late July, peak in mid-August, and decline through September, which is consistent with the expected temporal evolution of toxic blooms (Figure 2a). Second, high-probability detections are concentrated in nearshore shallow waters, particularly near the Maumee River mouth.

4. Conclusion

We paired 2024–2025 in situ MC measurements with PACE OCI hyperspectral surface reflectance in the WLEB and showed that OCI spectra contain learnable surrogate signals associated with elevated MC. A bootstrapped Welch's *t*-test identified significant wavebands and metrics, and a nested cross-validated random forest detected high MC ($\geq 1 \mu\text{g L}^{-1}$) with 89.5% accuracy, an F1 score of 0.8, AUROC of 0.945, and AUPRC of 0.841. Feature-importance analysis indicated that pigment-informed spectral-shape metrics, along with predictors in the red-edge and near-UV regions, rank among the most informative features. The resulting spatiotemporal predictions were consistent with the regional and seasonal climatology of MC occurrence.

Although the matchup sample size is limited, the results demonstrate the potential of PACE OCI for routine synoptic screening of high-toxin risk in optically complex inland waters. Future work should expand multi-year matchups, distinguish particulate from dissolved MC to better resolve release dynamics, and further assess the information content of near-UV reflectance for MC detection. Incorporating hydrologic and optical variables (e.g., turbidity, CDOM) may improve robustness. Extending the approach to smaller inland waters may be enabled by higher-resolution missions such as Landsat Next.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

In situ Lake Erie data are available from NOAA NCEI (Chaffin, 2023; Cooperative Institute for Great Lakes Research et al., 2019). PACE OCI Level-2 surface reflectance (SFREFL) data are available from NASA's Ocean Biology Distributed Active Archive Center (OB.DAAC) (NASA Ocean Biology Processing Group, 2025).

Acknowledgments

We gratefully acknowledge funding support from the Legislative-Citizen Commission on Minnesota Resources (LCCMR, 2024-037). Additional funding was provided by the U.S. Army Engineer Research and Development Center through the Aquatic Nuisance Species Research Program's HAB Research & Development Initiative, Grant W912HZ24C0092.

References

- Bailey, S. W., & Werdell, P. J. (2006). A multi-sensor approach for the on-orbit validation of ocean color satellite data products. *Remote Sensing of Environment*, 102(1–2), 12–23. <https://doi.org/10.1016/j.rse.2006.01.015>
- Benjamini, Y., & Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B*, 57(1), 289–300. <https://doi.org/10.1111/j.2517-6161.1995.tb02031.x>
- Binding, C. E., Greenberg, T. A., & Bukata, R. P. (2013). The MERIS Maximum Chlorophyll Index; its merits and limitations for inland water algal bloom monitoring. *Journal of Great Lakes Research*, 39(S1), 100–107. <https://doi.org/10.1016/j.jglr.2013.04.005>
- Binding, C. E., Jerome, J. H., Bukata, R. P., & Booty, W. G. (2010). Suspended particulate matter in lake erie derived from modis aquatic colour imagery. *International Journal of Remote Sensing*, 31(19), 5239–5255. <https://doi.org/10.1080/01431160903302973>
- Bouaïcha, N., Miles, C. O., Beach, D. G., Labidi, Z., Nguyen-Quang, T., & Benayache, N. (2019). Structural diversity, characterization and toxicology of microcystins. *Toxins*, 11(12), 714. <https://doi.org/10.3390/toxins11120714>
- Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32. <https://doi.org/10.1023/A:1010933404324>
- Caplan, S., Franz, B., Ibrahim, A., & Bailey, S. (2025). Pace oci surface reflectance (sfrefl). NASA Algorithm Publication Tool. <https://doi.org/10.5067/MD7W4BQ3Z6J2>
- Carmichael, W. W. (2001). Health effects of toxin-producing cyanobacteria: “the cyanoHABs”. *Human and Ecological Risk Assessment: An International Journal*, 7(5), 1393–1407. <https://doi.org/10.1080/20018091095087>
- Cawley, G. C., & Talbot, N. L. C. (2010). On over-fitting in model selection and subsequent selection bias in performance evaluation. *Journal of Machine Learning Research*, 11, 2079–2107. Retrieved from <https://www.jmlr.org/papers/volume11/cawley10a/cawley10a.pdf>
- Chaffin, J. (2023). Total microcystins, chlorophyll, and other water quality data collected in lake erie from 2013-06-18 to 2024-10-22 [Dataset]. NOAA National Centers for Environmental Information (NCEI). <https://doi.org/10.25921/7c57-gf98>
- Clark, R. N., & Roush, T. L. (1984). Reflectance spectroscopy: Quantitative analysis techniques for remote sensing applications. *Journal of Geophysical Research*, 89(B7), 6329–6340. <https://doi.org/10.1029/JB089iB07p06329>
- Cooperative Institute for Great Lakes Research, University of Michigan, & NOAA Great Lakes Environmental Research Laboratory. (2019). Physical, chemical, and biological water quality monitoring data to support detection of harmful algal blooms (habs) in western lake erie, collected by the great lakes environmental research laboratory and the cooperative institute for great lakes research since 2012 [Dataset]. NOAA National Centers for Environmental Information. <https://doi.org/10.25921/11da-3x54>
- Davis, J., & Goadrich, M. (2006). The relationship between precision–recall and roc curves. In *Proceedings of the 23rd international conference on machine learning (icml '06)* (pp. 233–240). ACM. <https://doi.org/10.1145/1143844.1143874>
- Dekker, A. G., Hoogenboom, H. J., & Peters, S. W. M. (1993). The use of thematic mapper for the analysis of eutrophic lakes: A case study in the Netherlands. *International Journal of Remote Sensing*, 14(5), 799–821. <https://doi.org/10.1080/01431169308904379>
- Dierssen, H. M., Ackleson, S. G., Joyce, K. E., Hestir, E. L., Castagna, A., Lavender, S., & McManus, M. A. (2021). Living up to the hype of hyperspectral aquatic remote sensing: Science, resources and outlook. *Frontiers in Environmental Science*, 9, 649528. <https://doi.org/10.3389/fenvs.2021.649528>
- Dierssen, H. M., Gierach, M., Guild, L. S., Mannino, A., Salisbury, J., Schollaert Uz, S., et al. (2023). Synergies between NASA's hyperspectral aquatic missions PACE, GLIMR, and SBG: Opportunities for new science and applications. *JGR Biogeosciences*, 128(9), e2023JG007574. <https://doi.org/10.1029/2023JG007574>
- Efron, B., & Tibshirani, R. J. (1994). *An introduction to the bootstrap*. Chapman & Hall/CRC. <https://doi.org/10.1201/9780429246593>

- Feng, L., Wang, Y., Hou, X., Qin, B., Kutser, T., Qu, F., et al. (2024). Harmful algal blooms in inland waters. *Nature Reviews Earth & Environment*, 5(8), 631–644. <https://doi.org/10.1038/s43017-024-00578-2>
- Francy, D. S., Brady, A. M. G., Ecker, C. D., Graham, J. L., Stelzer, E. A., Struffolino, P., et al. (2016). Estimating microcystin levels at recreational sites in western lake erie and ohio. *Harmful Algae*, 58, 23–34. <https://doi.org/10.1016/j.hal.2016.07.003>
- Gitelson, A. A., Garbuzov, G., Szilagyi, F., Mittenzwey, K.-H., Karnieli, A., & Kaiser, A. (1993). Quantitative remote sensing methods for real-time monitoring of inland water quality. *International Journal of Remote Sensing*, 14(7), 1269–1295. <https://doi.org/10.1080/01431169308953956>
- Gons, H. J. (1999). Optical teledetection of chlorophyll *a* in turbid inland waters. *Environmental Science & Technology*, 33(7), 1127–1132. <https://doi.org/10.1021/es9809657>
- Gower, J., King, S., Borstad, G., & Brown, L. (2008). The importance of a band at 709 nm for interpreting water-leaving spectral radiance. *Canadian Journal of Remote Sensing*, 34(3), 287–295. <https://doi.org/10.5589/m08-022>
- Harke, M. J., Steffen, M. M., Gobler, C. J., Otten, T. G., Wilhelm, S. W., Wood, S. A., & Paerl, H. W. (2016). A review of the global ecology, genomics, and biogeography of the toxic cyanobacterium, microcystis spp. *Harmful Algae*, 54, 4–20. <https://doi.org/10.1016/j.hal.2015.12.007>
- Ho, J. C., Michalak, A. M., & Pahlevan, N. (2019). Widespread global increase in intense lake phytoplankton blooms since the 1980s. *Nature*, 574(7778), 667–670. <https://doi.org/10.1038/s41586-019-1648-7>
- Huisman, J., Codd, G. A., Paerl, H. W., Ibelings, B. W., Verspagen, J. M., & Visser, P. M. (2018). Cyanobacterial blooms. *Nature Reviews Microbiology*, 16(8), 471–483. <https://doi.org/10.1038/s41579-018-0040-1>
- Hunter, P. D., Matthews, M. W., Kutser, T., & Tyler, A. N. (2016). Remote sensing of cyanobacterial blooms in inland, coastal, and ocean waters. In J. Meriluoto, L. Spoof, & G. A. Codd (Eds.), *Handbook of cyanobacterial monitoring and cyanotoxin analysis* (pp. 89–99). John Wiley & Sons. <https://doi.org/10.1002/9781119068761.ch9>
- International Joint Commission (IJC). (2014). A balanced diet for lake erie: Reducing phosphorus loadings and harmful algal blooms (Technical Report). *International Joint Commission*. Retrieved from <https://ijc.org/en/balanced-diet-lake-erie-reducing-phosphorus-loadings-and-harmful-algal-blooms>
- Janatian, N., Raudsepp, U., Broomandi, P., Fickas, K., Olli, K., Heimovaara, T., et al. (2025). A review on remote-sensing-based harmful cyanobacterial bloom monitoring services. *Remote Sensing Applications: Society and Environment*, 37, 101488. <https://doi.org/10.1016/j.rsase.2025.101488>
- Kaplan, A., Harel, M., Kaplan-Levy, R. N., Hadas, O., Sukenik, A., & Dittmann, E. (2012). The languages spoken in the water body (or the biological role of cyanobacterial toxins). *Frontiers in Microbiology*, 3, 138. <https://doi.org/10.3389/fmicb.2012.00138>
- Kim, J.-H. (2009). Estimating classification error rate: Repeated cross-validation, repeated hold-out and bootstrap. *Computational Statistics & Data Analysis*, 53(11), 3735–3745. <https://doi.org/10.1016/j.csda.2009.04.009>
- Kokaly, R. F., & Clark, R. N. (1999). Spectroscopic determination of leaf biochemistry using band-depth analysis of absorption features and stepwise multiple linear regression. *Remote Sensing of Environment*, 67(3), 267–287. [https://doi.org/10.1016/S0034-4257\(98\)00084-4](https://doi.org/10.1016/S0034-4257(98)00084-4)
- Kutser, T. (2004). Quantitative detection of chlorophyll in cyanobacterial blooms by satellite remote sensing. *Limnology and Oceanography*, 49(6), 2179–2189. <https://doi.org/10.4319/lo.2004.49.6.2179>
- Makarewicz, J. C. (1993). Phytoplankton biomass and species composition in lake erie, 1970 to 1987. *Journal of Great Lakes Research*, 19(2), 258–274. [https://doi.org/10.1016/S0380-1330\(93\)71216-2](https://doi.org/10.1016/S0380-1330(93)71216-2)
- Mannino, A., Novak, M. G., Nelson, N. B., Belz, M., Berthon, J.-F., Blough, N. V., et al. (2019). Measurement protocol of absorption by chromophoric dissolved organic matter (cdom) and other dissolved materials. In A. Mannino & M. G. Novak (Eds.), *Inherent optical property measurements and protocols: Absorption coefficient*. Dartmouth, NS, Canada: International Ocean Colour Coordinating Group (IOCCG). Retrieved from https://ioccg.org/wp-content/uploads/2019/10/cdom_abs_protocol_public_draft-19oct-2019-sm.pdf
- Matthews, M. W. (2011). A current review of empirical procedures of remote sensing in inland and near-coastal transitional waters. *International Journal of Remote Sensing*, 32(21), 6855–6899. <https://doi.org/10.1080/01431161.2010.512947>
- Matthews, M. W., & Odermatt, D. (2015). Improved algorithm for routine monitoring of Cyanobacteria and eutrophication in inland and near-coastal waters. *Remote Sensing of Environment*, 156, 374–382. <https://doi.org/10.1016/j.rse.2014.10.010>
- Millie, D. F., Fahnenstiel, G. L., Bressie, J. D., Pigg, R. J., Rediske, R. R., Klarer, D. M., et al. (2009). Late-summer phytoplankton in western lake erie (laurentian great lakes): Bloom distributions, toxicity, and environmental influences. *Aquatic Ecology*, 43(4), 915–934. <https://doi.org/10.1007/s10452-009-9238-7>
- Mishra, S., Stumpf, R. P., Schaeffer, B., Werdell, P. J., Loftin, K. A., & Meredith, A. (2021). Evaluation of a satellite-based Cyanobacteria bloom detection algorithm using field-measured microcystin data. *Science of The Total Environment*, 774, 145462. <https://doi.org/10.1016/j.scitotenv.2021.145462>
- Mobley, C. D. (Ed.) (2022). *The oceanic optics book*. Dartmouth, NS, Canada: International Ocean Colour Coordinating Group (IOCCG). <https://doi.org/10.25607/OBP-1710>
- Moore, T. S., Mouw, C. B., Sullivan, J. M., Twardowski, M. S., Burtner, A. M., Ciochetto, A. B., et al. (2017). Bio-optical properties of Cyanobacteria blooms in western lake erie. *Frontiers in Marine Science*, 4, 300. <https://doi.org/10.3389/fmars.2017.00300>
- NASA Ocean Biology Processing Group. (2025). PACE OCI Level-2 Regional Surface Reflectance Data, version 3.1 [Dataset]. *NASA Ocean Biology Distributed Active Archive Center (OB.DAAC)*. <https://doi.org/10.5067/PACE/OCI/L2/SFREFL/3.1>
- Nishizawa, T., Asayama, M., Fujii, K., Harada, K.-i., & Shirai, M. (1999). Genetic analysis of the peptide synthetase genes for a cyclic heptapeptide microcystin in microcystis spp. *The Journal of Biochemistry*, 126(3), 520–529. <https://doi.org/10.1093/oxfordjournals.jbchem.a022481>
- Olmanson, L. G., Bauer, M. E., & Brezonik, P. L. (2008). A 20-year landsat water clarity census of minnesota's 10,000 lakes. *Remote Sensing of Environment*, 112(11), 4086–4097. <https://doi.org/10.1016/j.rse.2007.12.013>
- Olmanson, L. G., Brezonik, P. L., & Bauer, M. E. (2011). Evaluation of medium to low resolution satellite imagery for regional lake water quality assessments. *Water Resources Research*, 47(9), W09515. <https://doi.org/10.1029/2011WR011005>
- Paerl, H. W., & Otten, T. G. (2013). Harmful cyanobacterial blooms: Causes, consequences, and controls. *Microbial Ecology*, 65(4), 995–1010. <https://doi.org/10.1007/s00248-012-0159-y>
- Saito, T., & Rehmsmeier, M. (2015). The precision-recall plot is more informative than the roc plot when evaluating binary classifiers on imbalanced datasets. *PLoS One*, 10(3), e0118432. <https://doi.org/10.1371/journal.pone.0118432>
- Sauer, M. J., Roesler, C. S., Werdell, P. J., & Barnard, A. (2012). Under the hood of satellite empirical chlorophyll *a* algorithms: Revealing the dependencies of maximum band ratio algorithms on inherent optical properties. *Optics Express*, 20(19), 20920–20933. <https://doi.org/10.1364/OE.20.020920>
- Sayers, M. J., Grimm, A. G., Shuchman, R. A., Bosse, K. R., Fahnenstiel, G. L., Ruberg, S. A., & Leshkevich, G. A. (2019). Satellite monitoring of harmful algal blooms in the Western Basin of Lake Erie: A 20-year time-series. *Journal of Great Lakes Research*, 45(3), 508–521. <https://doi.org/10.1016/j.jglr.2019.01.005>

- Simis, S. G. H., Peters, S. W. M., & Gons, H. J. (2005). Remote sensing of the cyanobacterial pigment phycocyanin in turbid inland water. *Limnology and Oceanography*, 50(1), 237–245. <https://doi.org/10.4319/lo.2005.50.1.0237>
- Stumpf, R. P., Davis, T. W., Wynne, T. T., Graham, J. L., Loftin, K. A., Johengen, T. H., et al. (2016). Challenges for mapping cyanotoxin patterns from remote sensing of Cyanobacteria. *Harmful Algae*, 54, 160–173. <https://doi.org/10.1016/j.hal.2016.01.005>
- Stumpf, R. P., Wynne, T. T., Baker, D. B., & Fahnenstiel, G. L. (2012). Interannual variability of cyanobacterial blooms in lake erie. *PLoS One*, 7(8), e42444. <https://doi.org/10.1371/journal.pone.0042444>
- Sukenik, A., & Kaplan, A. (2021). Cyanobacterial harmful algal blooms in aquatic ecosystems: A comprehensive outlook on current and emerging mitigation and control approaches. *Microorganisms*, 9(7), 1472. <https://doi.org/10.3390/microorganisms9071472>
- Vincent, R. K., Qin, X., McKay, R. M. L., Miner, J., Czajkowski, K. P., Savino, J., & Bridgeman, T. (2004). Phycocyanin detection from landsat tm data for mapping cyanobacterial blooms in lake erie. *Remote Sensing of Environment*, 89(3), 381–392. <https://doi.org/10.1016/j.rse.2003.10.014>
- Wang, S., & Qin, B. (2025). Application of optical remote sensing in harmful algal blooms in lakes: A review. *Remote Sensing*, 17(8), 1381. <https://doi.org/10.3390/rs17081381>
- Werdell, P. J., Behrenfeld, M. J., Bontempi, P. S., Boss, E., Cairns, B., Davis, G. T., et al. (2019). The plankton, aerosol, cloud, ocean ecosystem mission: Status, science, advances. *Bulletin of the American Meteorological Society*, 100(9), 1775–1794. <https://doi.org/10.1175/BAMS-D-18-0056.1>
- WHO. (2003). *Cyanobacterial toxins: Microcystin-Lr in drinking-water (Tech. Rep.)*. World Health Organization. <https://www.who.int/docs/default-source/wash-documents/wash-chemicals/cyanobacterial-toxins-background-document.pdf>
- Wiedner, C., Visser, P. M., Fastner, J., Metcalf, J. S., Codd, G. A., & Mur, L. R. (2003). Effects of light on the microcystin content of microcystis strain pcc 7806. *Applied and Environmental Microbiology*, 69(3), 1475–1481. <https://doi.org/10.1128/AEM.69.3.1475-1481.2003>
- Wynne, T. T., Stumpf, R. P., Tomlinson, M. C., & Dyble, J. (2010). Characterizing a cyanobacterial bloom in western lake erie using satellite imagery and meteorological data. *Limnology and Oceanography*, 55(5), 2025–2036. <https://doi.org/10.4319/lo.2010.55.5.2025>
- Wynne, T. T., Stumpf, R. P., Tomlinson, M. C., Warner, R. A., Tester, P. A., Dyble, J., & Fahnenstiel, G. L. (2008). Relating spectral shape to cyanobacterial blooms in the laurentian great lakes. *International Journal of Remote Sensing*, 29(12), 3665–3672. <https://doi.org/10.1080/01431160802007640>
- Yacobi, Y. Z., Köhler, J., Leunert, F., & Gitelson, A. (2015). Phycocyanin-specific absorption coefficient: Eliminating the effect of chlorophylls absorption. *Limnology and Oceanography: Methods*, 13(4), 157–168. <https://doi.org/10.1002/lom3.10015>
- Zhang, F., Hu, C., Shum, C. K., Liang, S., & Lee, J. (2017). Satellite remote sensing of drinking water intakes in lake erie for Cyanobacteria population using two modis-based indicators as a potential tool for toxin tracking. *Frontiers in Marine Science*, 4, 124. <https://doi.org/10.3389/fmars.2017.00124>
- Zolfaghari, K., Pahlevan, N., Binding, C., Gurlin, D., Simis, S. G. H., Ruiz Verdú, A., et al. (2021). Impact of spectral resolution on quantifying Cyanobacteria in lakes and reservoirs: A machine-learning assessment. *IEEE Transactions on Geoscience and Remote Sensing*, 60, 1–20. <https://doi.org/10.1109/TGRS.2021.3114635>